

# CELL TRANSPORT – DIFFUSION & OSMOSIS

## KEY VOCABULARY

**Diffusion:** net movement of particles from high to low concentration.

**Osmosis:** diffusion of water across a partially permeable membrane.

**Active transport:** moving substances against the gradient using energy.

**Concentration gradient:** the difference in concentration between two areas.

## COMMON MISCONCEPTIONS

~~— Diffusion needs energy from respiration.~~

✓ Diffusion is passive — it needs no energy.

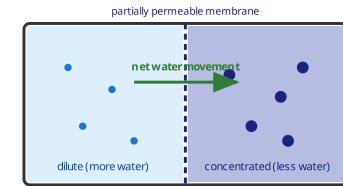
~~— Osmosis is the movement of any substance.~~

✓ Osmosis is the movement of water only.

~~— Particles stop moving once balanced.~~

✓ Particles still move, but there is no net change.

## OSMOSIS ACROSS A MEMBRANE



Water moves from a **dilute** solution to a **concentrated** one.

### 1 DEFINE DIFFUSION

Describe what is meant by **diffusion**. [2 marks]

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### 2 OSMOSIS

Explain how osmosis is different from diffusion. [2 marks]

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### 3 RATE OF DIFFUSION

State **three factors** that increase the rate of diffusion. [3 marks]

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### 4 REQUIRED PRACTICAL

Potato cylinders are placed in sugar solution and lose mass. **Explain why**. [2 marks]

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### 5 PERCENTAGE CHANGE

A cylinder changes from 5.0 g to 4.0 g. **Calculate** the percentage change in mass. [2 marks]

Answer: \_\_\_\_\_ %

### 6 ACTIVE TRANSPORT

Complete the sentence using the word bank. [2 marks]

Active transport moves substances \_\_\_\_\_ the concentration gradient and requires \_\_\_\_\_.

against

energy

with

no energy

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